

Abdikafar Omar

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EDUCATION

Duke University

Expected May 2027

Bachelor of Science in Computer Science, Concentration in Artificial Intelligence and Machine Learning

Durham, NC

- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics and Proofs, Computer Architecture, Statistical Data Science, Introduction to Databases, Design and Analysis of Algorithms

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, R, SQL

Frameworks/Libraries: React, Next.js, LangChain, Streamlit, Pandas, NumPy, scikit-learn, NLTK, Matplotlib, PyTorch, Plotly

Tools/Platforms: Git, Docker, Linux, Node.js, ChromaDB, OpenAI API, REST APIs, UiPath, UiPath Orchestrator

RELEVANT EXPERIENCE

AI Automation Intern

May 2026 – Aug 2026

Great American Insurance Group

Cincinnati, OH

- Engineered field-level ground-truth datasets for aviation and crop-insurance document pipelines, defining expected outputs used to test and evaluate AI extraction behavior.
- Designed and iterated extraction prompts for ambiguous aviation terminology, distinguishing passenger counts such as “pax” from crew-related values and diagnosing failures across edge cases.
- Performed end-to-end testing of a UiPath email-ingestion workflow that routed unread submissions from a shared inbox into crop and agriculture queues, validating bot pickup and downstream processing through UiPath Orchestrator.

Software Engineering Intern

May 2024 – Aug 2024

Duke Office of Information Technology (Code+ Program)

Durham, NC

- Architected and implemented the core backend of the Scholars@Duke Opportunities Finder, an AI-powered platform matching university researchers with relevant funding and collaboration opportunities.
- Built an end-to-end semantic-retrieval pipeline with LangChain and ChromaDB, transforming faculty profiles and opportunity metadata into vector embeddings and ranking matches through contextual similarity search.
- Integrated metadata ingestion, embedding generation, vector indexing, and result retrieval into the full-stack application while converting researcher feedback into tested product functionality.

PROJECTS

Scholars@Duke Opportunities Finder

React, Python, LangChain, ChromaDB

- Continued developing the Scholars@Duke Opportunities Finder beyond its initial release, maintaining the backend that connects faculty with relevant funding and collaboration opportunities.
- Improved data ingestion, vector indexing, and semantic search to keep opportunity recommendations current, relevant, and reliable.

Black Box Auditor

Python, Streamlit, scikit-learn, Plotly, GitHub Actions

- Architected a reproducible multi-model evaluation engine that applies seven adversarial prompt mutations under identical temperature and token settings through offline and OpenAI-compatible providers.
- Implemented transparent safety metrics, TF-IDF/K-means behavioral clustering, CSV/JSON/PDF exports, and 12 automated tests plus a GitHub Actions CI workflow.

EcoMind AI Carbon Footprint Explorer

Python, Streamlit, Pandas, Plotly, GitHub Actions

- Engineered a transparent scenario-analysis engine that converts model parameters and training tokens into FLOPs, IT and facility energy consumption, and operational CO₂e using configurable hardware efficiency, data-center overhead, and grid intensity.
- Built validated batch analysis for up to 500 models, three-scenario sensitivity comparisons, interactive visualizations, CSV/JSON/PDF exports, and 14 automated tests with GitHub Actions.

LEADERSHIP & INVOLVEMENT

Co-President

Aug 2026 – Present

Duke Black Muslim Coalition (BMC_o)

Durham, NC

- Lead organizational strategy, funding proposals, event execution, and partnerships with campus and regional organizations serving Black Muslim students.